

And, explain multiple inheritance in PHP with example.

19. Design the following form in HTML and display the value in the result box after calculating basic arithmetic operations based on user input with use of JavaScript.

Basic Arithmetic Functions				
Operand	Operand			
ADD	SUB	MUL	DIV	MOD
Result				

20. Design following form in HTML and write corresponding PHP and SQL code to store the user's values after satisfying following validation rules using JavaScript.

- a) Length of full name up to 20 character (*Normal JavaScript Validation*).
- b) E-Mail address must be valid email address (*Regular Expression Validation*).
- c) Username must be start with string followed by number (*Regular Expression Validation*).
- d) Password length must be more than eight character (*Regular Expression Validation*).

Note: Assume your own database and database connectivity constrains if necessary.

Register	
Full name*	Email*
Username*	Password*
<i>* Required fields</i>	
SUBMIT	

Best of Luck



Babarmahal, Kathmandu
Pre-board Examination-2082

Bachelor in Computer Application
Course Title: Software Engineering
Code No: CACS 253
Semester: IV

Full Marks: 60
Pass Marks: 24
Time: 3 Hours

Group B

Attempt any SIX questions.

[6x5=30]

11. What is a waterfall model? Describe the activities of waterfall model and also mention its drawbacks. [2+3]
12. What is CASE? Explain the importance of CASE tools in software development life cycle. [2+3]
13. Define modularization? Differentiate between cohesion and coupling with suitable example. [2+3]
14. What are programming languages? Explain various software development tools. [2+3]
15. Discuss COCOMO model in cost estimation of the software in detail. [5]
16. What do you mean by verification and validation? Differentiate between Black Box testing and White Box testing. [2+3]
17. Explain activities in software quality assurance with example. [5]

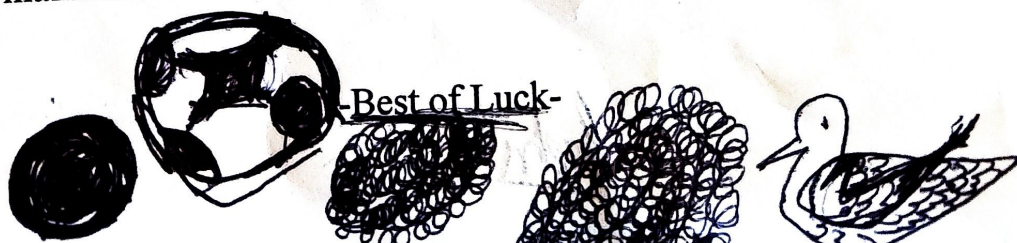
Group C

Attempt any TWO questions.

[2x10=20]

18. Explain challenges faced by software engineering along with professional and ethical responsibility. [10]
19. What is Capability Maturity Model? Explain its levels along with key process area, Differentiate between alpha and beta testing. [2+4+4]
20. What do you mean by software maintenance? Explain the different types of maintenance? Explain reengineering process in detail. [2+4+4]

Best of Luck-



T1	T2	T3
READ (A)		
	READ (B)	
		READ (B)
	WRITE (B)	
WRITE(A)		
Commit		
		WRITE (A)
	READ (A)	
	WRITE (A)	
		WRITE(C)
	WRITE(C)	
	Commit	
		Commit

19. PATIENT(Pid,Pname ,Age, Gender)

DOCTOR(Did,Dname, Age, Gender, speciality)

TEST(Tid,Tname,PRICE)

CONSULTS(Pid, Did)

ORDERS(Tid, Did)

Write Sql Queries for following.

- Find out name of patients who is a heart patient.
 - Find out total no of patients of Dr. Ram
 - What is the name of doctor whose name starts from A and ends with A?
 - Give discount to female patients by 10% on price of their tests.
 - Display name of doctor who orders x-ray.
20. Explain basic steps used in query processing with help of a diagram.
What is cost incurred doing selection using file scan using Binary Search?

THE END



Babarmahal, Kathmandu
Pre-board Examination-2082

Bachelor in Computer Application
Course Title: Database Mgmt. System
Code No: CACS 255
Semester: IV

Full Marks: 60
Pass Marks: 24
Time: 3 Hours

Group B

Attempt any Six questions.

[6×5=30]

11. What is functional Dependency in DBMS? State and Explain 1NF and 2NF with the help of an example.
12. State and explain various Transaction States with help of a diagram.
13. Define Weak Entity. Explain completeness constraints with example.
14. Product (Pid, Pname, Price, Categories). Write Relation algebra expression to
 - a. Insert one new row in given relation.
 - b. Increase price by 10% of product in drinks category.
 - c. Find out average price of all categories.
15. Define Trigger and explain simple trigger with help of example, syntax and appropriate query.
16. Consider a university database for the scheduling of classroom for final exam. This database could be modeled as entity set **exam** (course_no, section-number, room_number, time). And additional entity sets could be defined, along with relationship set to place some of the attributes of the exam entity set as **Course** (name, department, c_number), **Section** (s_number, enrollment, department) and **Room** (r_number, capacity, building). Now draw an E-R diagram illustrating the use of all entity set listed.
17. Define Privileges and its types. Grant privileges to create and modify table and insert and retrieve data in student table to users Ram, Shyam and Sita.

Group C

Attempt any Two questions.

[10×2=20]

18. State and explain two phase locking protocol, conservative 2pl and rigorous 2pl with example. Also find out if the following schedule is conflict serializable. (6+4)

- a) FCFS
- b) SRTF
- c) Priority (Preemptive)
- d) RR

Process Burst Time Arrival Time Priority

P1 5 0 2

P2 15 1 3

P3 10 2 1

20. Define the term seek time and rotational delay in disk scheduling.

Suppose that the disk has 100 cylinders, numbered 0 to 199. The drive is currently serving a request at cylinder 53. The queue of pending request, in FIFO order is: 98, 183, 37, 122, 14, 124, 65, 67. Starting from current head position, what is total distance (in cylinders) that the disk arm moves to satisfy all pending request for each of following disk scheduling algorithms?

- a) FCFS
- b) SSTF
- c) SCAN
- d) LOOK

All the Best



Babarmahal, Kathmandu
Pre-board Examination-2082

Bachelor in Computer Application
Course Title: Numerical Methods
Code No: CACS 252
Semester: IV

Full Marks: 60
Pass Marks: 24
Time: 3 Hours

Group B

Attempt any SIX questions.

[6x5=30]

11. What are the importance of Numerical Method in the field of science and engineering? Define error in Numerical method.
12. What are the limitations of newton -Raphson method? Using this method, the find the real root of the equation $\cos x - 3x + 1 = 0$, correct to four decimal places.
13. Solve using partial pivoting method
 $2x + y + z - 2u = -10$
 $4x + 2z + u = 8$
 $3x + 2y + 2z = 7$
 $x + 3y + 2z - u = -5$
14. Estimate the value of $y(4.5)$ using Newton difference interpolation formula.

X	1	2	3	4	5
Y	0.5	2	4.5	8	12.5

15. Evaluate $\int_{-1}^2 e^{-x^2} dx$ using gaussian 3-point formula.
16. Derive the general Newton-Cote quadrature formula and use it obtain Simpson's 1/3 formula.
17. Write a pseudo-code to solve first order differential equation using RK-4 method.

Group C

Attempt any TWO Questions.

[2x10=10]

18. Estimate $y(4)$ using cubic spline interpolation technique from the following data.



New
Edition

X	3	5	7	9	11
Y	6	9	12	9	6

19. Write the differences between initial value problem and boundary value problem. Solve the following boundary value problem using shooting method by dividing the interval into four sub intervals.

$$Y'' + 3Y' - Y = \cos x \quad y(0) = 2 \text{ and } y(2) = 3$$

20. Derive the Laplace equation.

Solve the discretized form of Laplace's equation, $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, for $u(x, y)$ defined within the domain of $0 \leq x \leq 1$ and $0 \leq y \leq 1$, given the boundary conditions (I) $u(x, 0) = 1$ (II) $u(x, 1) = 2$ (III) $u(0, y) = 1$ (IV) $u(1, y) = 2$

Best of luck



Babarmahal, Kathmandu
Pre-board Examination-2082

Bachelor in Computer Application

Course Title: Operating System

Code No: CACS 251

Semester: IV

Full Marks: 60

Pass Marks: 24

Time: 3 Hours

Group B

Attempt any Six questions.

[6 x 5 = 30]

11. Explain operating system as a resource manager and as an extended machine
12. Define two operations of semaphore. Write the necessary conditions for deadlock.
13. Explain different set of operations that can be performed on processes.
14. What is DMA? Explain the working mechanism of DMA.
15. Consider the page references 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1. Find the number of page fault using FIFO and LRU with 3-page frame.
16. What is the task of disk controller? Compare paging and Segmentation.
17. What do you mean by one-time password authentication? How worms are different from virus?

Group C

Attempt any TWO questions.

[2x10=20]

18. What is Coalescing and Compaction in memory management?
Differentiate between internal and external fragmentation? Consider six memory partitions of size 200 KB, 400 KB, 600 KB, 500 KB, 300 KB and 250 KB. These partitions need to be allocated to four processes of sizes 357 KB, 210 KB, 468 KB and 491 KB in that order. Perform the allocation of processes using:
 - a) Best Fit Algorithm
 - b) Worst Fit Algorithm
19. What are the CPU scheduling criteria? For the processes listed in the following table, draw the Gantt chart illustrating their execution and calculate average waiting time and average turnaround time where the time quantum is 2 unit using:

Babarmahal, Kathmandu
Pre-board Examination-2082

Bachelor in Computer Application
Course Title: Scripting Language
Code No: CACS 254
Semester: IV

Full Marks: 60
Pass Marks: 24
Time: 3 Hours

Group B

[6×5 = 30]

Attempt any SIX questions.

11. What is CMS and why we use? Write the steps to create submenu in WordPress or Joomla.
12. Why AJAX is required in web development? Write a sample program to synchronize data between JavaScript and PHP.
13. What is dialog box? Explain different dialog boxes with suitable example in JavaScript.
14. What is the use of jQuery? Write sample program to show and hide the certain <div> with use of jQuery.
15. What is PHP? Write PHP statement of MySQL database connection string with status check.
16. Change background color of <p> tag on button click and append as childNode with use of JS.
17. Describe Functions, Array Types and Class-Object with an example in PHP.

Group C

Attempt any TWO questions.

[2×10 = 20]

18. Write an Object-Oriented PHP Program to implement the concept of inheritance in considering with following class diagram with use of constructor for 20 students.

